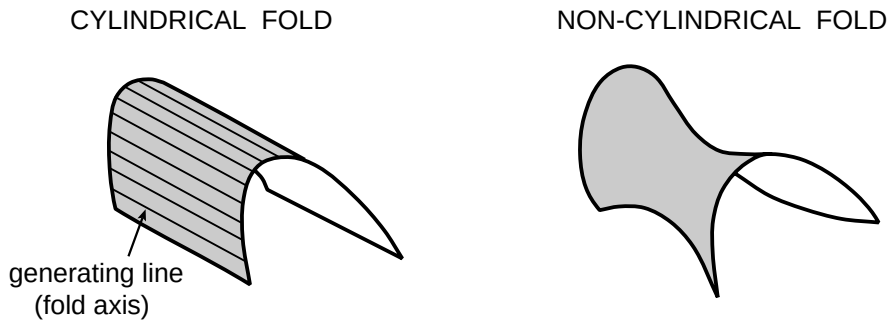
SPECIFYING THE GEOMETRY OF FOLDS (1)

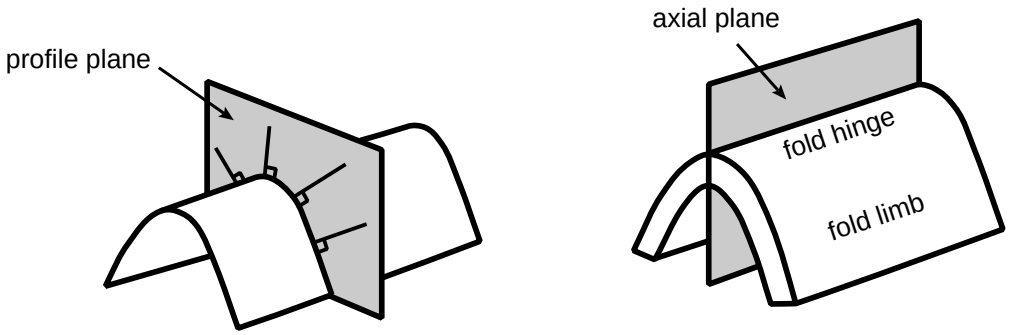
CYLINDRICAL AND NON-CYLINDRICAL FOLDS

• A curved surface, the shape of which can be generated by taking a straight line and moving it whilst keeping it parallel to itself, is called a cylindrically folded surface. The line capable of generating the surface is referred to as the ***fold axis***.

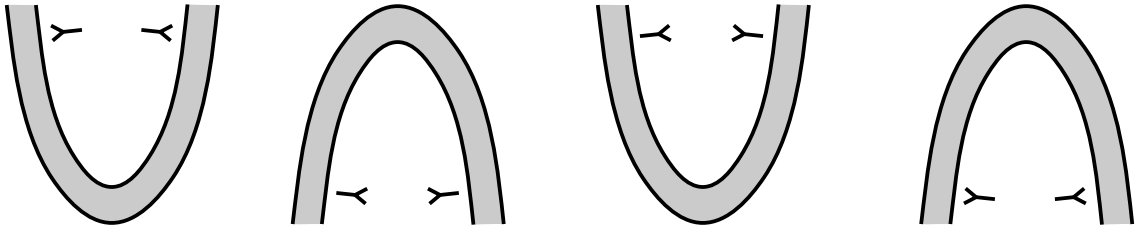
• An important property of cylindrical folds is that the fold shape on parallel cross-section planes, whatever their orientation, is constant.



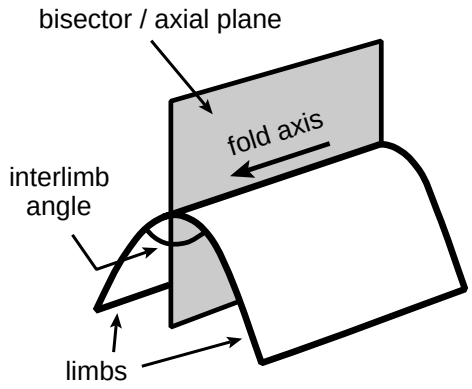
BASIC GEOMETRICAL FEATURES OF A FOLD



• A folded surface which closes downwards is referred to as a ***synform***; one which closes upwards is an ***antiform***. A fold in which the youngest rocks are in the core is referred to as a ***syncline***; one in which the oldest rocks are in the core is an ***anticline***.



SPECIFYING THE GEOMETRY OF FOLDS (2)



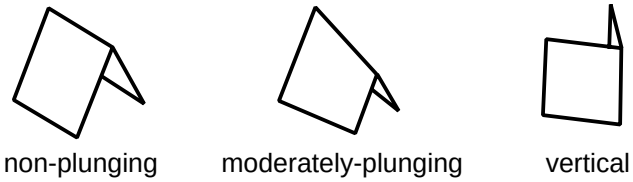
• Apart from providing some indication of the scale of the fold, and some basic observations about its shape (*e.g.*, antiform/synform, anticline/syncline), there are four things which are required to specify the geometry of a cylindrical fold fully:

(1) the orientation (plunge/plunge direction) of the fold axis;
(2) the orientation (strike/dip) of the axial plane;
(3) representative orientations of each limb of the fold;
(4) the interlimb angle.

• Depending on the plunge of the fold axis, the size of the interlimb angle, and the dip of the axial plane, the following adjectives are widely applied to describe the geometry of folds.

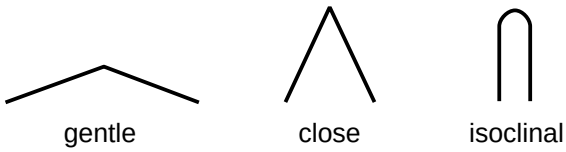
Plunge of fold axis

0°-10°	non-plunging
10°-30°	gently-plunging
30°-60°	moderately-plunging
60°-80°	steeply-plunging
80°-90°	vertical



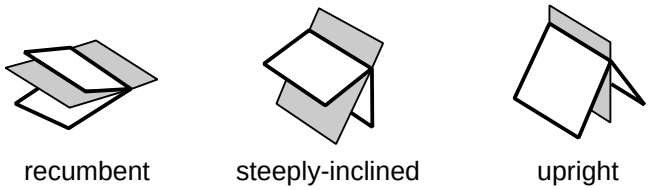
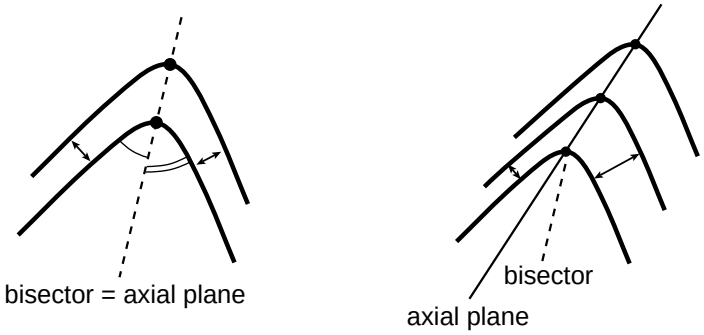
Interlimb angle

180°-120°	gentle
120°-70°	open
70°-30°	close
30°-0°	tight
0°	isoclinal



Dip of axial plane

0°-10°	recumbent
10°-30°	gently-inclined
30°-60°	moderately-inclined
60°-80°	steeply-inclined
80°-90°	upright



Strictly, the axial plane is the plane which contains the fold axis and the line in the profile plane which joins the points of maximum curvature on adjacent beds. If the beds on each limb are of the same thickness the axial plane bisects the interlimb angle; if the beds on each limb are of different thickness, it does not.